GA-SFS: A Two-Stage Hybrid Algorithm for Feature Selection in Biomedical Data

Brian Bales
Department of Computer Science
California State University, Northridge
California, USA
brian.bales.253@my.csun.edu

Monitha Davuluri
Department of Computer Science
California State University, Northridge
California, USA
monitha.davuluri.606@my.csun.edu

Rashida Hasan
Department of Computer Science
California State University, Northridge
California, USA
rashida.hasan@csun.edu

***Abstract*—In biomedical data analysis, feature selection is crucial, particularly for high-dimensional datasets where redundant or irrelevant features might affect model performance. Biomedical datasets introduce additional challenges, such as high dimensionality, small sample sizes, and complex data structures. Addressing these issues often requires hybrid, adaptive, or ensemble-based algorithms that can improve model robustness and interpretability. To meet these demands, this paper introduces GA-SFS, a two-stage hybrid algorithm that combines Genetic Algorithms (GA) with Sequential Forward Selection (SFS) to select informative features in complex biomedical datasets effectively. Experimental results on multiple datasets show that GA-SFS outperforms traditional feature selection methods in classification accuracy.**

***Index Terms*—Feature Selection, Genetic Algorithm, Sequential Forward Selection, Biomedical Datasets, Hybrid Model.**

I. Introduction

Machine learning and deep learning approaches have been applied widely in biomedical fields such as disease detection, medical imaging, genomics for identifying genetic markers, drug discovery for predicting compound efficacy, and personalized medicine for tailoring treatments [1]. Biomedical data encompass a wide range of information generated from biological and clinical sources, including genomic sequences, medical imaging, electronic health records, and wearable health sensors. These datasets are often high-dimensional, with vast amounts of data points for each individual, and can contain features spanning genetic, molecular, and clinical domains. In addition, these data often involve a large number of features collected from a relatively small number of samples [2]. Many of these features may be irrelevant or redundant, complicating the analysis and leading to potential issues such as overfitting. When non-informative features dominate, they can obscure the underlying patterns that are crucial for accurate predictions or classifications [3].

Feature selection in machine learning refers to the process of selecting a subset of relevant and important features from a larger set of available features to improve the performance of a model [4]. Feature selection approaches in biomedical data can be classified into three main types: supervised, unsupervised, and semi-supervised [5]. However, in this work, we focus on supervised feature selection, which involves choosing features based on their ability to predict the target variable. The supervised feature selection can be further classified into three categories: filter, wrapper, and embedded methods [3]. Filter methods assess feature relevance independently using statistical tests or correlation measures, making them efficient for high-dimensional datasets but unable to capture feature interactions. Wrapper methods consider feature interactions by evaluating subsets with a predictive model, often improving performance. Embedded methods integrate feature selection during model training, balancing efficiency and predictive power.

Common filtering approaches in the biomedical domain include AUC score, kolmogorov–smirnov test, relief score, pearson correlation coefficient, t-test, mutual information [1], [5]–[9]. The embedded algorithms for feature selection in the biomedical domain include embedded neural network, and SVM-based method [10], [11]. The main disadvantage of these embedded feature selection methods lies in their dependency on the chosen machine learning model. In addition, most of these focus on specific applications, for example, cancer classification data. Many works focused on wrapper methods including iterative feature removal and optimization-based algorithms such as genetic algorithm [12], [13]. With the rise of neural networks, several solutions have emerged that focus on leveraging neural network-based approaches for feature selection [10], [14]. One of the primary challenges of using deep learning algorithms for feature selection in biomedical data is the issue of small sample sizes with high-dimensional data. Biomedical datasets often involve a large number of features, such as gene expressions, protein levels, or imaging data, but the number of available samples (patients or cases) is relatively small. This leads to the small sample size, high-dimensional (SSHD) dilemma, in which there are substantially more features than there are training cases [15].

Recently, hybrid models have gained significant traction in feature selection for the biomedical domain due to their ability to combine the strengths of multiple feature selection techniques, thereby improving the overall performance and robustness of predictive models [1]. The fast SFFS-based hybrid algorithm selects 20-25% features at each iteration and combines with SVM to select features [16]. Authors of [17] proposed a hybrid algorithm based on maximum relevance-minimum redundancy and simulated annealing algorithm.

979-8-3315-0769-5/25/$31.00 ©2025 IEEE

Some of the hybrid work focused on specific applications such as feature selection for cardiovascular disease or heart disease dataset [18], [19].

In this work, we proposed a two-stage hybrid algorithm GA-SFS based on a genetic algorithm and a sequential forward selection process. The Genetic Algorithm (GA) is an optimization-based approach inspired by natural selection. It refines feature subsets through selection, crossover, and mutation, identifying combinations that maximize model performance. This is particularly useful for high-dimensional biomedical datasets, where testing all possible feature combinations is computationally infeasible. In contrast, the Sequential Floating Forward Selection (SFFS) algorithm takes a simpler, more direct approach, dynamically adding and removing features to find the optimal subset. In our hybrid approach, combining the Genetic Algorithm (GA) and Sequential Floating Forward Selection (SFFS) leverages the strengths of both optimization and dynamic feature adjustment. GA efficiently explores high-dimensional spaces for optimal feature subsets, while SFFS fine-tunes the selection, ensuring both computational efficiency and improved model performance in biomedical data classification. The contribution of our work can be summarized as follows:

- We introduced a hybrid algorithm, GA-SFS, based on the filter and wrapper approaches. We employed a genetic algorithm combined with Sequential Forward Selection to select relevant features.
- We conducted a thorough experimental evaluation using diverse datasets, comparing the proposed algorithm's performance with the state-of-the-art method.
- We conducted evaluations to demonstrate the impact of feature selection by comparing the performance of the model with and without feature selection,

The remainder of this paper is organized as follows: Section II details the methodology of the proposed algorithm. In Section III, we present the datasets employed and discuss the experimental results, along with a comparison to state-of-the-art methods. Lastly, Section V concludes the paper, highlighting the main findings and proposing avenues for future research.

II. Methodology

In this section, we defined the problem of feature selection in biomedical datasets and presented a hybrid approach to address it. We combine the strengths of Genetic Algorithms (GA) and Sequential Forward Selection (SFS) to create the GA-SFS hybrid model.

A. Problem Statement

Definition 1: Biomedical Data: Biomedical data refers to high-dimensional information from various medical and biological sources, including genetic data, clinical records, and medical imaging. The term "high-dimensional" is defined as datasets containing a large number of features (e.g., genes, biomarkers, or imaging attributes) in relation to the relatively small number of available samples.

Definition 2: Feature Selection in Supervised Learning: Given a dataset with a high-dimensional feature space, feature selection refers to the process of identifying and selecting a subset of the most relevant features from the larger set to improve model performance and reduce computational complexity. The problem of feature selection in supervised learning can be formally defined as:

$$S^* = \arg\max_{S \subseteq F} A(S) \tag{1}$$

Where: S^* is the optimal subset of selected features, S is a candidate subset of features, F is the set of all available features, and $A(S)$ is a performance metric (such as accuracy, F1 score, or cross-validation error) evaluated on the selected feature subset S.

B. Proposed Algorithm

Our hybrid approach consists of two modules: The genetic algorithm module to choose the candidate feature set and the Sequential forward selection module to select the final optimal feature subset. By combining GA and SFS in the GA-SFS hybrid model, we aim to leverage the global search capabilities of GA with the local refinement efficiency of SFS to improve feature selection performance.

1) Genetic Algorithm: Our feature selection approach GA-SFS leverages a Genetic Algorithm (GA) to systematically find an optimal subset of features from high-dimensional biomedical data. This algorithm operates through the following sequence of steps:

- Initialization: We begin by randomly generating an initial population of chromosomes, each representing a unique combination of features. To encourage variety, each chromosome includes at least three features, covering a broad range of potential feature combinations from the start.
- Fitness Evaluation: We evaluated each chromosome's fitness using a Support Vector Machine (SVM) classifier with five-fold stratified cross-validation to ensure consistent class distribution across folds. The F1 score is chosen as the fitness metric, reflecting the need to balance precision and recall in biomedical datasets, which often face class imbalance challenges. Given a chromosome's feature subset S, the fitness $F(S)$ is evaluated by training an SVM model with five-fold Stratified K-Fold cross-validation. For each fold i, the F1 score is computed as:

$$F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{2}$$

The overall fitness of the chromosome's feature subset S is the average F1 score across all folds:

$$Fitness(S) = \frac{1}{k} \sum_{i=1}^{k} F1_i \tag{3}$$

where k is the number of folds. This approach balances precision and recall across folds, ideal for managing class imbalances in biomedical datasets.

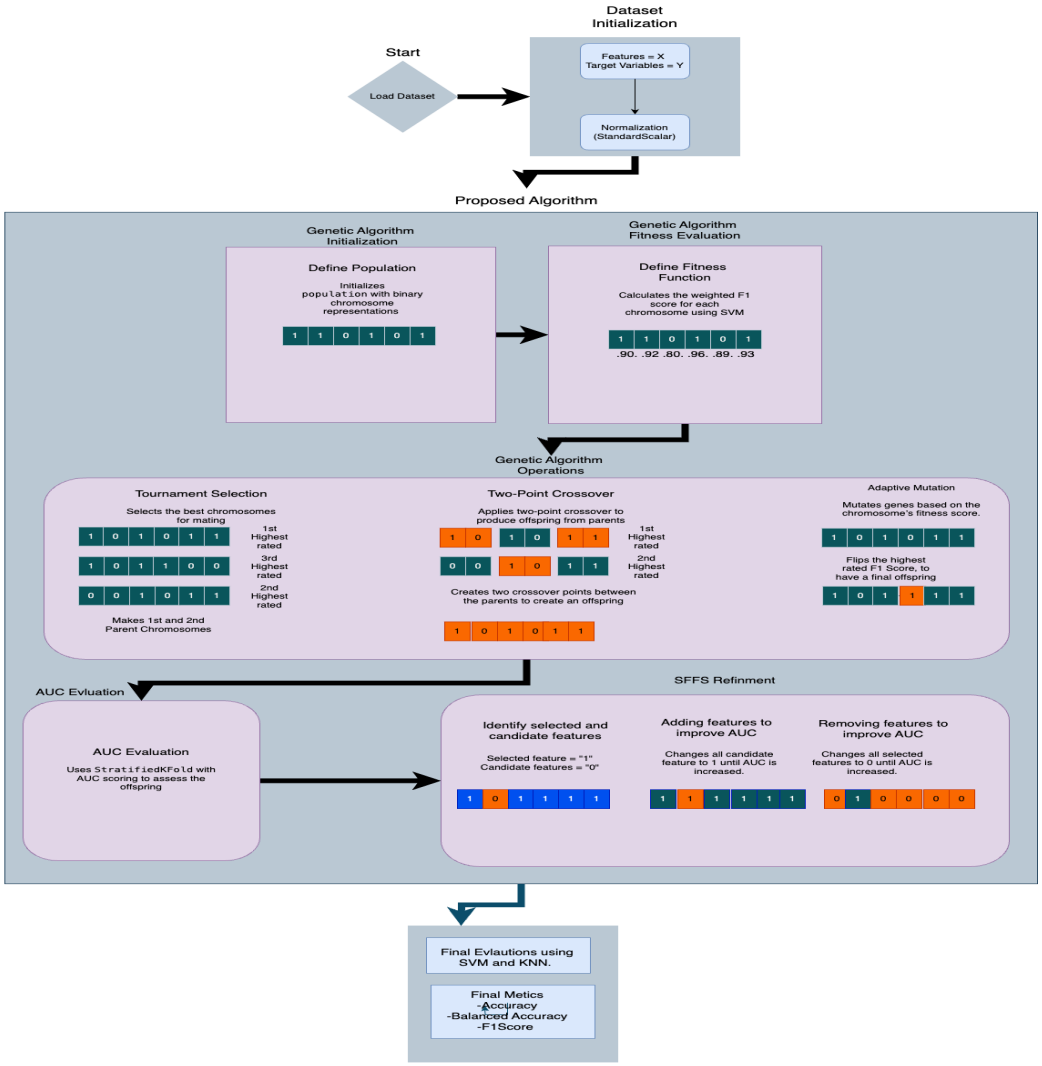

Fig. 1: The General Framework of GA-SFS

- Selection (Tournament Selection): Chromosomes are selected as parents through a tournament selection strategy, where the fittest chromosome from a random subset is chosen. This selection process introduces selective pressure, increasing the likelihood of choosing chromosomes with higher fitness while maintaining diversity in the population. The process of tournament selection can be mathematically represented as:

$$\text{Parent} = \arg\max_{i \in T} f(x_i) \tag{4}$$

  Where: T is a randomly chosen subset of individuals from the population, $f(x_i)$ is the fitness function evaluated for individual i in the subset T, and $\arg\max$ indicates selecting the individual with the highest fitness in the subset T as the parent.
- Crossover (Two-Point Crossover): Selected parent chromosomes undergo a two-point crossover, where two sections of the chromosomes are swapped. This allows offspring chromosomes to explore new feature combinations by blending the best features from each parent. Consider two parent chromosomes P_1 and P_2, each consisting of a set of features. The two-point crossover involves selecting two crossover points, say at positions cp_1 and cp_2, where $cp_1 < cp_2$, and swapping the feature subsets between these two positions. For each offspring O_i: Select two crossover points cp_1 and cp_2 randomly and create O_i by combining segments from two parents. The crossover operation can be mathematically represented as:

$$O_i[1 : cp_1] = \text{Parent1}[1 : cp_1] \tag{5}$$

$$O_i[cp_1 : cp_2] = \text{Parent2}[cp_1 : cp_2] \tag{6}$$

$$O_i[cp_2 : F] = \text{Parent1}[cp_2 : F] \tag{7}$$

- Adaptive Mutation: To maintain diversity, we use an adaptive mutation rate, meaning chromosomes with lower

fitness are given higher mutation rates. This encourages exploration in parts of the feature space that may have been overlooked, helping the algorithm avoid getting stuck in local optima. For each gene g in O_i, the adaptive mutation is defined as:

$$\text{adaptive_rate} = \mu \times \left(1 - \frac{\text{Fitness}(O_i)}{\max \text{Fitness}(\text{Pop}_g)}\right) \quad (8)$$

where

– pop_g represents the gene g in the population

2) SFFS: The second component of our hybrid model is Sequential Forward Feature Selection (SFFS) where we iteratively add and remove features to find the optimal subset. In this process, we use the AUC-ROC score to evaluate the performance of the selected feature subset. The AUC-ROC score serves as a robust measure of a model's capability to differentiate between positive and negative classes, making it highly applicable to biomedical datasets.

Let F represent the set of all available features, and let S^* denote the optimal subset of features. The goal is to iteratively select a subset of features S^* that maximizes the AUC-ROC score. The optimization function can be written as

$$S^* = \arg\max_{S \subseteq F} \text{AUC-ROC}(S) \quad (9)$$

Where: S^* is the optimal subset of features, AUC-ROC(S) is the AUC-ROC score calculated for the feature subset S, and F is the set of all available features.

The process of SFFS with AUC-ROC is as follows:

- **Initialization:** Start with an empty set of selected features $S^* = \emptyset$.
- **Iteration:** At each step, add the feature $f_i \in S \setminus S^*$ that maximizes the AUC-ROC score:

$$\begin{aligned} S^*(t+1) &= S^*(t) \cup \{f_i\}, \\ f_i = \arg \max_{f \in S \setminus S^*(t)} &\text{AUC-ROC}(S^*(t) \cup \{f\}) \end{aligned} \quad (10)$$

- **Selection:** After adding a new feature, remove features that reduce the AUC-ROC score:

$$\begin{aligned} S^*(t+1) &= S^*(t) \setminus \{f_j\}, \\ f_j = \arg \min_{f \in S^*(t)} &\text{AUC-ROC}(S^(t) \setminus \{f_j\}) \end{aligned} \quad (11)$$

3) Hyrbird Module: Our hybrid module combines the strengths of Genetic Algorithms (GA) and Sequential Forward Floating Selection (SFFS) to optimize feature selection. Initially, the GA is employed to explore the feature space, using evolutionary processes such as selection, crossover, and mutation to generate diverse feature subsets. Once the GA has identified potential subsets, the process is refined by SFFS, which iteratively adds or removes features based on their impact on the AUC-ROC score. The general framework of our approach is shown in Fig1

III. Experiments and Results

We evaluate the performance of our algorithm from the following three aspects: (i) is feature selection effective in biomedical datasets? (ii) can our algorithm perform well on binary and multiclass datasets? (iii) can we outperform state-of-the-art algorithms? In this section, we provide details about our dataset and the performance of our algorithm. All the experiments were conducted in Google Colab, utilizing the Tesla T4 GPU for accelerated computation, with Python 3.10.

A. Dataset Description

We collect the dataset from the UCI machine learning repository [20], [21], and [22]. The datasets wdbc, wpbc and spect are collected from the UCI machine learning repository. The SBRCT dataset is collected from [22] and the other two datasets (prostate and colon) are collected from [21]. To demonstrate the superiority of our algorithm, we have selected both binary and multi-class datasets. We also include datasets with a range of class distributions, spanning from slightly imbalanced(for example prostate) to highly imbalanced scenarios(for example spect). TableI shows the detailed characteristics of our dataset.

B. Experimental setup

We evaluated our proposed algorithm GA-SFS with two popular classification model K-Nearest Neighbor (KNN) and Support Vector Machine (SVM). Since biomedical datasets are typically highly imbalanced, we use stratified k-fold cross-validation with k=5 to preserve the original class distribution in each fold, ensuring a more accurate and reliable evaluation. We use the following parameter setup for our Genetic Algorithm: number of generations: 15, population size=7, number of parents mating=7, tournament size=3, base mutation ate=0.05. To evaluate our model's performance, we use three metrics: accuracy, F1 score, and balanced accuracy. The reason for choosing the F1 score and balanced accuracy is that biomedical datasets are often imbalanced, so we want to ensure that all classes are correctly classified without favoring the majority class.

C. Model Performance: With and Without Feature Selection

We present an experiment demonstrating that selecting relevant features significantly enhances model performance. Table II highlights the results, showcasing performance improvements across two different evaluation classifiers: KNN and SVM. The number of selected features is substantially reduced across six datasets, with reductions reaching nearly 50% or more in the Prostate, SBRCT, and Colon datasets. However, the original feature size for WDBC, WPBC, and SPECT is relatively smaller compared to the other three datasets. Despite this, the reduction in the number of features remains significant. In the case of KNN evaluation, our algorithm outperforms across all datasets when compared with no feature selection. There is a noticeable improvement in KNN performance with feature selection, particularly in the Prostate dataset (from 0.75 to 0.92), the SBRCT dataset

TABLE I: Dataset Characteristics

Dataset	Features	Samples	Class	Class Distribution
Prostate	6033	102	2	class 1: 59%, class 2: 49%
SBRCT	2308	83	4	class 1: 34.9% , class 2: 13.3%, class 3: 21.7%, class 4: 30.1%
Colon	2000	62	2	class 1: 35%, class 2: 65%
WDBC	30	569	2	class 1: 62.8%, class 2:37.2%
WPBC	33	198	2	class 1:65.5%, class 2:34.5%
SPECT	44	267	2	class 1: 79%, class 2: 21%

TABLE II: Model Performance: With and Without Feature Selection

Dataset	Original Features	Selected Features	KNN		SVM	
			Without FS	FS(GA-SFS)	Without FS	FS(GA-SFS)
Prostate	6033	2948	0.75	0.92	0.87	0.98
SBRCT	2308	1144	0.81	0.97	1.00	0.96
Colon	2000	996	0.76	0.82	0.74	0.90
WDBC	30	13	0.92	0.97	0.90	0.98
WPBC	33	16	0.92	0.97	0.90	0.98
SPECT	44	22	0.74	0.95	0.95	0.99

(0.81 to 0.97), and the SPECT dataset (from 0.74 to 0.95). For SVM evaluation, our algorithm shows superior results in 6 out of 6 datasets. However, for the SPECT dataset, the changes are minimal. One possible reason could be the SVM algorithm itself is robust to redundant features. In both KNN and SVM evaluations, both large and small datasets show significant performance improvements with our feature selection approach.

D. Performance Analysis: Proposed Approach GA-SFS vs State-of-the-art algorithm

We compare our algorithm with the state-of-the-art Fast SFFS algorithm [16]. The Fast SFFS algorithm combines filter and wrapper methods for feature selection. It randomly selects an initial subset of features, typically around 20-25% of the total, which yields a higher AUC than using all the features. The candidate subset generated by the filter approach is then evaluated using k-fold cross-validation with a support vector machine (SVM), where a user-defined classification margin is applied as a wrapper [18].

We present the comparative analysis of our method, GA-SFS, with the state-of-the-art algorithm in TableIII, TableIV, and TableV. It is evident that our GA-SFS consistently outperforms state-of-the-art algorithms across six datasets in terms of accuracy, F1 score, and balanced accuracy. Our algorithm shows better performance compared to the state-of-the-art in both evaluation classifiers (KNN and SVM). For datasets like Prostate, SBRCT, Colon, and WPBC, GA-SFS boosts performance for both KNN and SVM classifiers, demonstrating its effectiveness in selecting the most relevant features. While its performance on the WDBC dataset is competitive in terms of accuracy, GA-SFS outperforms the state-of-the-art in terms of F1 score and balanced accuracy. Our algorithm performs well even on multi-class datasets. For instance, in the SBRCT dataset, which has four classes, GA-SFS significantly improves performance for both KNN and SVM. The F1 score for KNN increases from 0.85 to 0.95, and for SVM, it increases from 0.88 to 0.99.

IV. Efficiency and Practical Application

The proposed algorithm's time complexity is influenced by chromosome refinement steps and fitness evaluation function, resulting in a time complexity of $O(G \cdot P \cdot F + K)$, where G is the number of generations, P is the population size, F is the number of features, and K is the complexity of model evaluation. The GA-SFS hybrid algorithm has real-world potential, especially in scenarios like early cancer detection. For instance, when working with gene expression data, which often contains thousands of features, the algorithm can identify the most critical genetic markers linked to specific cancer types. It will help narrow down the data to the most relevant features, it enables researchers and clinicians to build more accurate predictive models for identifying cancer at an early stage. This targeted approach not only helps diagnostics but also reduces computational complexity, making it practical for real-world biomedical applications.

V. Conclusion

This study presented a hybrid GA-SFFS approach for feature selection in high-dimensional biomedical datasets. In our two-stage hybrid method, we utilized the genetic algorithm and the sequential forward selection to select the relevant feature. The proposed method demonstrated superior performance over an advanced SFFS-based state-of-the-art method across multiple metrics, including accuracy, balanced accuracy, and F1 score. A limitation of our algorithm is its high computational cost, as it adds one feature at a time during the Sequential Feature Selection (SFS) process. In our future work, we plan to incorporate an advanced version of SFS and introduce adaptive tournament selection within the genetic algorithm to reduce time complexity. We plan to compare our algorithm with faster algorithms to evaluate the efficiency of our algorithm. Additionally, we aim to adapt the algorithm for unsupervised feature selection, expanding its applicability to a broader range of tasks.

TABLE III: GA-SFS vs State-of-the-art algorithm (Accuracy)

Dataset	KNN		SVM	
	GA-SFS	**State-of-the-art algorithm**	**GA-SFS**	**State-of-the-art algorithm**
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.97	0.83	1.00	0.89
Colon	0.82	0.62	0.90	0.67
WDBC	0.97	0.96	0.98	0.99
WPBC	0.97	0.95	0.98	0.90
SPECT	0.95	0.77	0.99	0.92

TABLE IV: GA-SFS vs State-of-the-art algorithm (F1-Score)

Dataset	KNN		SVM	
	GA-SFS	**State-of-the-art algorithm**	**GA-SFS**	**State-of-the-art algorithm**
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.96	0.85	1.00	0.89
Colon	0.80	0.62	0.90	0.67
WDBC	0.97	0.94	0.98	0.96
WPBC	0.97	0.94	0.98	0.94
SPECT	0.95	0.85	0.99	0.88

TABLE V: GA-SFS vs State-of-the-art algorithm (Blanaced Accuracy)

Dataset	KNN		SVM	
	GA-SFS	**State-of-the-art algorithm**	**GA-SFS**	**State-of-the-art algorithm**
Prostate	0.92	0.87	0.98	0.84
SBRCT	0.94	0.86	1.00	0.89
Colon	0.76	0.68	0.88	0.72
WDBC	0.96	0.94	0.97	0.96
WPBC	0.96	0.95	0.98	0.95
SPECT	0.92	0.64	0.98	0.64

REFERENCES

[1] Y. Peng, Z. Wu, and J. Jiang, "A novel feature selection approach for biomedical data classification," *Journal of Biomedical Informatics*, vol. 43, no. 1, pp. 15–23, 2010.

[2] E. Pashaei and E. Pashaei, "An efficient binary chimp optimization algorithm for feature selection in biomedical data classification," *Neural Computing and Applications*, vol. 34, no. 8, pp. 6427–6451, 2022.

[3] P. Dhal and C. Azad, "A comprehensive survey on feature selection in the various fields of machine learning," *Applied Intelligence*, vol. 52, no. 4, pp. 4543–4581, 2022.

[4] J. Li, Z. Li, L. Song, and P. S. Yu, "Feature selection: A data perspective," *ACM Computing Surveys (CSUR)*, vol. 50, no. 6, pp. 1-45, 2017, doi: 10.1145/3136625.

[5] J. Labory, E. Njomgue-Fotso, and S. Bottini, "Benchmarking feature selection and feature extraction methods to improve the performances of machine-learning algorithms for patient classification using metabolomics biomedical data," *Computational and Structural Biotechnology Journal*, vol. 23, pp. 1274–1287, 2024.

[6] R. J. Urbanowicz *et al.*, "Relief-based feature selection: Introduction and review," *Journal of Biomedical Informatics*, vol. 85, pp. 189–203, 2018.

[7] L. Goh, Q. Song, and N. Kasabov, "A novel feature selection method to improve classification of gene expression data," in *Proc. 2nd Conf. Asia-Pacific Bioinformatics*, vol. 29, pp. 119–126, 2004.

[8] C. Chu *et al.*, "Does feature selection improve classification accuracy? Impact of sample size and feature selection on classification using anatomical magnetic resonance images," *Neuroimage*, vol. 60, no. 1, pp. 59–70, 2012.

[9] A. K. Tiwari *et al.*, "Hybrid similarity relation based mutual information for feature selection in intuitionistic fuzzy rough framework and its applications," *Scientific Reports*, vol. 14, no. 1, p. 5958, 2024.

[10] A. K. Naik and V. Kuppili, "An embedded feature selection method based on generalized classifier neural network for cancer classification," *Computers in Biology and Medicine*, vol. 168, p. 107677, 2024.

[11] P. El Kafrawy, M. S. R. S. S. Ahmed, H. M. Elsayed, and M. A. Elhoseny, "An efficient SVM-based feature selection model for cancer classification using high-dimensional microarray data," *IEEE Access*, vol. 9, pp. 155353-155369, 2021, doi: 10.1109/ACCESS.2021.3117061.

[12] M. R. Alnowami, F. A. Abolaban, and E. Taha, "A wrapper-based feature selection approach to investigate potential biomarkers for early detection of breast cancer," *Journal of Radiation Research and Applied Sciences*, vol. 15, no. 1, pp. 104–110, 2022.

[13] E. A. Alhenawi *et al.*, "Feature selection methods on gene expression microarray data for cancer classification: A systematic review," *Computers in Biology and Medicine*, vol. 140, p. 105051, 2022.

[14] Z. Chen, W. Liu, X. Li, P. Zhang, and Q. Wang, "Feature selection may improve deep neural networks for the bioinformatics problems," *Bioinformatics*, vol. 36, no. 5, pp. 1542–1552, 2020, doi: 10.1093/bioinformatics/btz823.

[15] S. Gao, Y. He, H. Li, W. Xu, Q. Zhang, and J. Sun, "RIFS2D: A two-dimensional version of a randomly restarted incremental feature selection algorithm with an application for detecting low-ranked biomarkers," *Computers in Biology and Medicine*, vol. 133, p. 104405, 2021, doi: 10.1016/j.compbiomed.2021.104405.

[16] F. Shirbani and H. Soltanian Zadeh, "Fast SFFS-based algorithm for feature selection in biomedical datasets," *AUT Journal of Electrical Engineering*, vol. 45, no. 2, pp. 43–56, 2013.

[17] E. Pashaei and E. Pashaei, "Hybrid binary arithmetic optimization algorithm with simulated annealing for feature selection in high-dimensional biomedical data," *The Journal of Supercomputing*, vol. 78, no. 13, pp. 15598-15637, 2022, doi: 10.1007/s11227-022-04586-3.

[18] Pavithra, V., and V. Jayalakshmi, "Hybrid feature selection technique for prediction of cardiovascular diseases," *Materials Today: Proceedings*, vol. 81, pp. 336-340, 2023.

[19] Doppala, Bhanu Prakash, et al., "A hybrid machine learning approach to identify coronary diseases using feature selection mechanism on heart disease dataset," *Distributed and Parallel Databases*, pp. 1-20, 2023.

[20] M. Kelly, R. Longjohn, and K. Nottingham, "The UCI Machine Learning Repository," Available: https://archive.ics.uci.edu.

[21] J. Li, K. Cheng, S. Wang, F. Morstatter, R. Trevino, J. Tang, and H. Liu, "Feature selection: A data perspective," *ACM Computing Surveys (CSUR)*, vol. 50, no. 6, pp. 1–45, 2017.

[22] J. Khan, J. S. Wei, M. Ringnér, L. H. Saal, M. Ladanyi, F. Westermann, F. Berthold, M.